Final Project Design Phase

Computer Science - Week 11

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For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I expect to use most concepts such as loops, conditionals, lists, and classes. I want to improve my code's modularity by being able to reuse certain functions all the time whenever needed.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The easiest part will be UI design because it seems the simplest. The hardest part will be implementing the reminders because it will have to remember the data as you travel to other pages.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The most time and effort would be the reminders and I think it will take about 1 hour to

fully get it right. I would like to implement the homepage and settings pages first.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions might be video games and watching youtube. Resources may include drafter documentation, python, and LLM. I might use previous hw assignments to help me with this project. I don't have any questions.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Home Page

Budget Buddy

Home
Bills
Budgeting
Settings

Bills
Name
Price
Paid

Name
Price
Unpaid

Budget
$ ~~~~.00

After subtracting the paid bills, this is your leftover amount

Bills page

+

Name: "~~~"
Price: $ ~~.00
Paid: PAID

Budgeting page

Enter your budget
$ ~~~.00

Settings Page
Dark Mode ☑

Finance Quiz page

None of this info is saved. This is solely to help you calculate how much you can pay toward your debt

1. Are you in debt? Yes
   If you don't have a debt skip the questions below NO
2. How much debt do you have? $ ~~.00
3. What is your yearly salary? $ ~~.00
4. What debt percentage would you like to use? ~~%

Complete

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**My site will help people with budgeting or with getting rid of debts.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	State bills: list[Bill] -To display your bills and can also be used to calculate leftover budget total message: str -This will be used to display the current status of the website budget: float -Keeps track of the current budget you inputted selected_bill: Bill -Helps with bill deletion or toggling between paid/unpaid
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Bill name: str -To know what the name of your bill is price: float -To know the price so it can calculate leftover budget paid: bool -Determines if it has been paid or not
Third Dataclass Your third dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Settings dark_mode: bool -If true then it makes the background dark
Routes (minimum 7) The names of your route functions	1. finance_quiz - displays the finance quiz on startup 2. home - shows bills and budget left (homepage) 3. go_to_bills - takes you to the bill page/displays

	bills 4. bill_form - shows template for bill creation 5. create_bill - creates it and adds it to the list of bills 6. delete_bill - deletes the bill 6. budget - takes you to the budgeting page* 7. settings - takes you to the settings page* 8. toggle_dark_mode - to switch between modes
Pages (minimum 5) A brief description of the pages in your site	A homepage(shows current bills/money left), a quiz page(Asks), a settings page, a bills(create/edit bills) page, and a budgeting(calculates your leftover money) page.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	User input1: textbox() Will collect the budget of the person User input2: textbox() Name of bill for bill creation User input3: textbox() Cost of bill to pay User input4: selectbox() Select if you are in debt or not
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Statement1: Will check if the bill has paid set to true to add to a new list Statement2: Will check if selected_bill exists in list and remove it via delete_bill Statement3: Will check if the user has pressed the complete button on the quiz to move onto the website's homepage
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it	This loop will iterate over the list of bills and get the total which will then be subtracted from the budget. Loop for filtering list of bills that would have only the paid ones.

is meaningful to the site	

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I am making a bill buddy drafter project. What things can I add to my state dataclass other than a list of bill

For your Bill Buddy project, your state dataclass can store more than just a list of bills. Think about what information your app needs to remember while the user interacts with it.

Here are some useful attributes you could add:

1. Suggested state attributes

Attribute	Data type	Purpose
`bills`	`list`	Stores all the user's bills
`total_monthly`	`float`	Tracks the total amount due each month
`paid_bills`	`list`	Keeps track of bills marked as paid
`unpaid_bills`	`list`	Keeps track of bills not yet paid
`selected_bill`	`str` or `None`	Stores the bill currently being viewed
`new_bill_name`	`str`	Stores the name entered into the form
`new_bill_amount`	`float`	Stores the amount entered into the form
`new_bill_due_date`	`str`	Stores the due date entered into the form

`filter_status`	`str`	Controls whether users see all, paid, or unpaid bills
`show_add_form`	`bool`	Controls whether the add-bill form is visible
`message`	`str`	Displays messages like "Bill successfully added!"

And what are some route functions I could make? I have a state, bill, and settings dataclass:

Bill:
name: str
price: float
paid: bool

State:
bills: list[Bill]
message: str
budget: float
Need one more

Settings:
dark_mode: bool

For Bill Buddy, I'd make your route functions focus on the main things a user would want to do: manage bills, track spending, and customize the app.

1. Route functions you could create

Route function	Purpose
`@route home`	Displays all bills and a spending summary
`@route add_bill`	Shows a form to add a new bill

`@route submit_bill`	Adds the submitted bill to the bills list
`@route delete_bill`	Removes a bill from the list
`@route mark_paid`	Changes a bill's paid status
`@route mark_unpaid`	Changes a bill back to unpaid
`@route budget`	Displays the user's budget and remaining money
`@route settings`	Shows app settings
`@route toggle_dark_mode`	Switches between dark and light mode

I need a different field for the State dataclass

I'd recommend adding:

```
selected_bill: Bill | None
```

Purpose: Keeps track of which bill the user has selected. This could be useful if they want to view details, edit, or delete a specific bill.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.